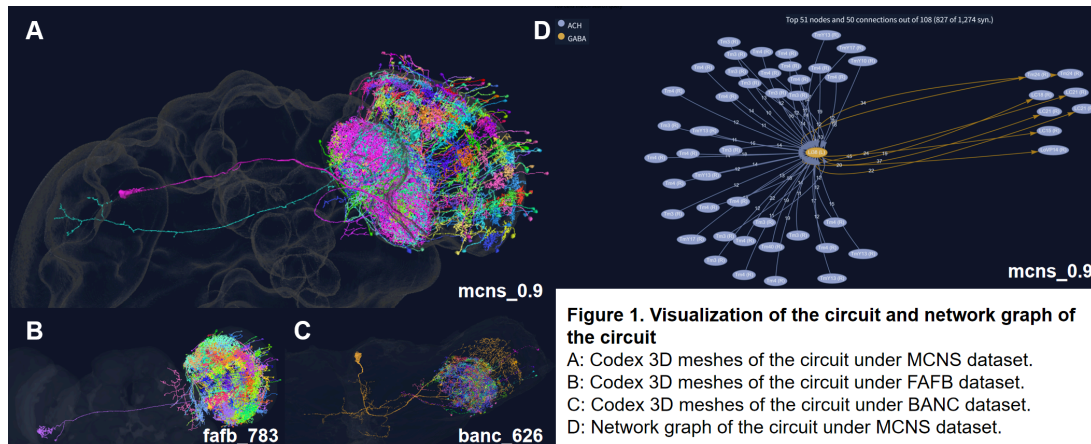


Putative Function of a Li38-Centered Visual Circuit

This study focuses on a conserved circuit of 109 neurons identified across the MCNS v0.9, BANC 626, and FAFB 783 datasets, with functional analyses based on annotations from MCNS v0.9.

Observations



This circuit is located at the right side of the brain (Fig 1A-C), dominated by optic-lobe intrinsic neurons (101/109 neurons) and is largely cholinergic (104/109 neurons). The most abundant neuron types are Tm3 (41 neurons) and Tm4 (32 neurons), with additional contributions from TmY13, TmY17, LC11, LC18, LC21, and a central lobula intrinsic neuron, Li38.

The strongest upstream inputs include L1/L2, Mi1, Tm3, Tm4, and T2a neurons. These cell types are associated with ON- and OFF-pathway visual processing. Network topology suggests that Li38 acts as a hub connecting many of the identified neurons (Fig 1D).

Interpretation and Biological Hypothesis

The circuit's composition points to a role in higher-order visual feature processing in the lobula. Abundant ON- and OFF-pathway neurons may enable integration of contrast and edge information, while TmY neurons could link multiple visual processing streams. The presence of object-related lobula output neurons (LC11, LC18, LC21) suggests this circuit contributes to detecting or representing small visual objects. Overall, I hypothesize that this Li38-centered network participates in object-feature extraction or small-target processing.

References:

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